Take-home task checklist

From The Data Science Interview Playbook by Mathew Kariuki. Use it alongside Chapter 12.

Before you open the data

☐ I can state the decision being made in one sentence.
☐ I know who the audience is (technical, leadership, both).
☐ I have listed the ambiguous terms and my assumption for each.
☐ I have emailed any blocking questions, with my default stated.
☐ I have a time plan that leaves an hour for writing.

Data checks

☐ Grain of every table confirmed (what is one row?).
☐ Keys are unique where they should be.
☐ Date range and any gaps checked.
☐ Missing values counted and their meaning understood.
☐ Totals reconciled with any figure given in the brief.
☐ Limitations noted (what the data cannot tell me).

Analysis

☐ Started with simple descriptive comparisons.
☐ Stated a baseline (base rate, simple rule, last period).
☐ Checked for obvious confounders and controlled for them.
☐ Put uncertainty on key numbers (confidence intervals, not just p-values).
☐ Translated technical results into business units (dollars, customers, hours).
☐ Did a back-of-envelope check of whether the answer matters.

Write-up (two pages or less)

☐ The first sentence answers the question.
☐ Three to five findings, each with a number.
☐ Limits stated honestly in a short paragraph.
☐ A clear recommendation and next steps.
☐ No code, regression tables or jargon in the summary.

Code

☐ README explains how to run it and which versions were used.
☐ Restarted and ran top to bottom without errors.
☐ No hard-coded paths to my own computer.
☐ Random seeds set.
☐ Unused exploration removed.

Before submitting

☐ Reread the brief. Did I answer the question they asked?
☐ Reread the summary as the busiest person on the panel.
☐ Prepared answers to: why this approach, weakest part, what next with more time.